INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR
MATHEMATICS CLUB
PROBLEM NUMBER: WP2501
DATE: OCTOBER 15, 2025
DUE DATE: NOVEMBER 01, 2025

WEEKLY PROBLEM

EXERCISE. Consider a 4×4 matrix whose entries are the integers 1 through 16 written in order left to right, top to bottom. Choose any set of four entries with the rule that exactly one chosen entry lies in each row and exactly one chosen entry lies in each column (equivalently: pick one entry from every row and the chosen column indices are all distinct). Prove that the sum of the four chosen entries is always 34. **Henceforth, generalize the result for an $N \times N$ matrix**.

Clarification for students: "Exactly one chosen entry in each row and each column" means no two chosen numbers share the same row or the same column. Such a choice is called a *transversal* (or a *permutation selection*) of the matrix.

Short illustrative examples: For the 4×4 matrix,

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

The circled numbers form one of the transversal selections. The sum of these entries $1 + 7 + 10 + 16 = 34$.
The different choices give the same total — you must prove why this always happens.

(Scan this to submit solution!)

Wish you a happy Diwali !!